

Order Flow Imbalance and Price Impact in CME Ether Futures

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Abstract

We study Order Flow Imbalance (Cont, Kukanov, and Stoikov, 2014) on CME Ether futures over February 2021 to August 2026. On 28,711 out-of-sample dollar-volume bars, an OLS regression of bar mid-quote change on contemporaneous OFI with Newey-West HAC standard errors yields $\hat{\beta}_0 = +0.215$ ($t = 58.1$, $R^2 = 0.30$), against $+0.300$ and $R^2 = 0.31$ in sample. The regression is contemporaneous, so it measures price impact rather than forecasting power: used on its own as a predictor, lagged OFI explains essentially none of the following bar’s move ($R^2 < 0.001$ out of sample). OFI is a strong descriptor of price formation on this contract; it is not, by itself, a trading signal.

1 Introduction

Order Flow Imbalance (OFI) aggregates best-bid and best-offer quote arrivals into a signed measure of top-of-book pressure. On liquid US equities, contemporaneous OFI explains roughly two-thirds of bar-level mid-quote variation (Cont, Kukanov, and Stoikov, 2014). The cryptocurrency evidence is thinner: Silantsev (2019) finds trade-flow imbalance dominates short-horizon prices on BitMEX XBT/USD, and Bieganowski and Ślepaczuk (2026) report OFI as the top SHAP feature in CatBoost ensembles on Binance Futures perpetuals. CME’s Ether contract sits apart from both: a regulated venue whose quoted spread, in basis points of price, is an order of magnitude wider than the equity tape’s.

This paper estimates the contemporaneous OFI–price relation on CME ETH futures from February 2021 to August 2026. The construction is the level-1 (best bid / best offer) form of Cont, Kukanov, and Stoikov on dollar-volume bars built from TBBO (trade-sampled top-of-book) events (Lopez de Prado, 2018).

2 Data

The panel consists of CME-cleared TBBO data for the front-month ETH futures contract, covering February 2021 through August 2026, with rolls handled by contract stitching.

Bars are constructed in dollar volume, with the threshold calibrated so that a median in-sample trading day produces about ten bars, and that threshold is held fixed when bars are constructed for the out-of-sample period.

For each bar the following quantities are computed:

- **ofi**: signed level-1 (best bid / best offer) order flow imbalance, following the construction of Cont, Kukanov, and Stoikov (2014). For each TBBO update i , let p_i^B, p_i^A denote the

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best-bid and best-ask quotes and q_i^B, q_i^A their sizes. The per-event OFI contribution is $e_i = e_i^B + e_i^A$, where

$$e_i^B = \begin{cases} +q_i^B & \text{if } p_i^B > p_{i-1}^B \\ -q_{i-1}^B & \text{if } p_i^B < p_{i-1}^B \\ q_i^B - q_{i-1}^B & \text{if } p_i^B = p_{i-1}^B \end{cases}, \quad e_i^A = \begin{cases} -q_i^A & \text{if } p_i^A < p_{i-1}^A \\ +q_{i-1}^A & \text{if } p_i^A > p_{i-1}^A \\ -(q_i^A - q_{i-1}^A) & \text{if } p_i^A = p_{i-1}^A. \end{cases}$$

The bar OFI is the unnormalised sum over all events in the bar, $O_t = \sum_{i \in \text{bar } t} e_i$.

- **mid_close**: the mid quote $(p^B + p^A)/2$ at the bar boundary.
- **d_mid**: the change in mid quote since the previous bar.

3 Signal-impact regression

Following Cont, Kukanov, and Stoikov (2014) and Silantyev (2019), bar mid-quote change is regressed on contemporaneous and one-bar lagged OFI:

$$\Delta p_t = \alpha + \beta_0 O_t + \beta_1 O_{t-1} + \varepsilon_t, \quad (1)$$

where $\Delta p_t = m_t - m_{t-1}$ is the change in mid-quote between bar boundaries and O_t is the bar OFI, so β_0 is in dollars of mid-quote change per contract of net imbalance. Standard errors are Newey-West HAC (Newey and West, 1987) with five lags. Table 1 reports the estimates on the full panel, the 2021-02 to 2023-12 in-sample (IS) window, and the 2024-01 to 2026-08 out-of-sample (OOS) window.

Window	n (bars)	$\hat{\beta}_0$	HAC $t(\beta_0)$	$\hat{\beta}_1$	HAC $t(\beta_1)$	R^2
Full sample	41,055	+0.247	+70.2	-0.027	-14.4	0.29
In-sample 2021–23	12,344	+0.300	+44.9	-0.025	-7.0	0.31
Out-of-sample 2024–26	28,711	+0.215	+58.1	-0.026	-13.3	0.30

Table 1: OLS, Newey-West HAC (five lags). $p < 10^{-6}$ on both coefficients in every window.

Both subsamples reject $\beta_0 = 0$ at $p < 10^{-6}$ (Figure 1). The IS and OOS R^2 are nearly identical (0.31 and 0.30), while the slope itself is less stable, moving from +0.300 in-sample to +0.215 out-of-sample. Both R^2 figures are below the two-thirds level of Cont et al. (2014) for liquid US equities.

This regression is contemporaneous: it measures how the mid quote moves together with the flow inside a bar (mean OOS bar duration ≈ 49 minutes), not whether flow forecasts the next bar. Used on its own as a predictor, lagged OFI explains essentially none of the following bar’s move ($R^2 < 0.001$ out-of-sample), so the content of Table 1 is price impact rather than forecastability. Whether the flow carries predictive content at horizons inside the bar is a question the bar regression cannot resolve, and answering it properly requires quote-level data.

4 Conclusion

The contemporaneous OFI–price relation is strong and stable on CME Ether futures: the explanatory power is nearly identical in and out of sample (R^2 of 0.31 and 0.30), and the slope remains large and precisely estimated throughout. It is a statement about price impact, not forecastability; lagged OFI alone explains essentially none of the next bar’s move, so the result describes how this market absorbs order flow rather than offering a trading signal.

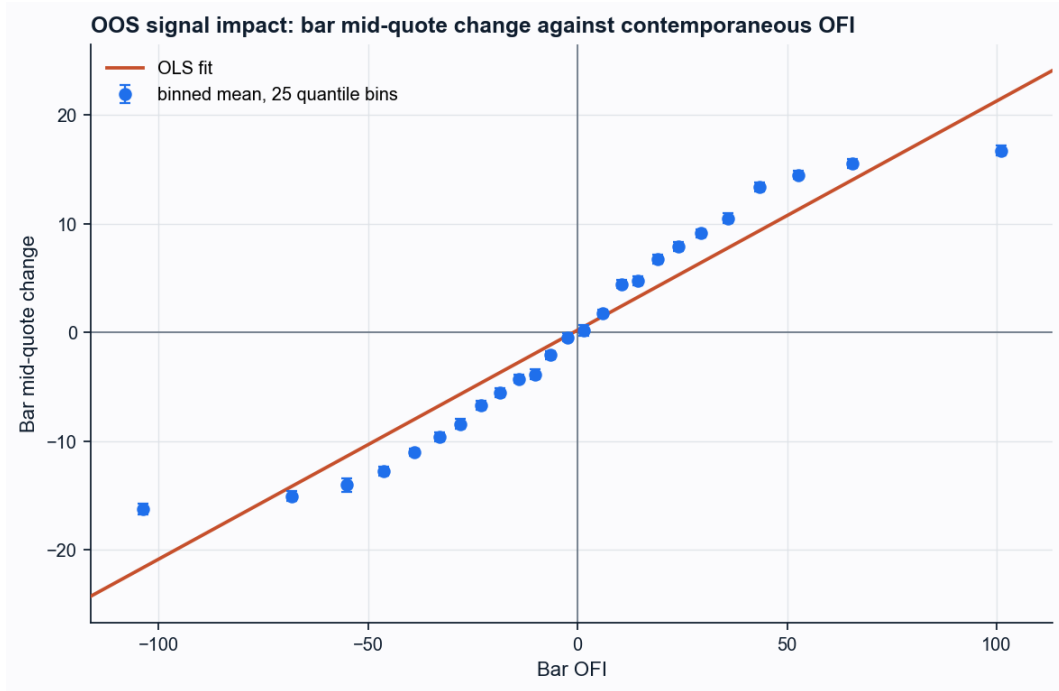


Figure 1: Out-of-sample bar mid-quote change against contemporaneous OFI, 25 quantile bins with ± 1 standard error and the OLS fit. The relationship is monotonic, with impact per unit of flow attenuating at extreme flows — both tail bins fall inside the straight-line fit; full estimates are in Table 1.

The tape bounds what can be measured. TBBO is trade-sampled, so the book is observed only when a trade prints, and it carries no depth beyond the touch. A quote-level feed would permit measuring the relation at horizons inside the bar, where any predictive content would live, and a proper treatment of execution; these are the natural extensions.

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